

# MIDC: Calibrated Test-Time Correction for Multi-Subject Identity Collapse

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## Abstract

Personalized image generation with multiple subjects suffers from *interaction-induced identity collapse*: as the number of subjects grows, models increasingly merge, swap, or drop identities, even when each subject is provided as a reference image. We introduce **MIDC**, a training-free, inference-time correction paradigm that treats a *decomposed* verifier as a structured diagnostic signal and routes correction to the most deficient facet. Given a candidate image, a decomposed evaluator scores three facets—existence, appearance, and interaction—and we convert each facet score into a standardized deficit against a calibrated per-facet baseline. A *calibrated routing* mechanism then selects which facet to correct and which action to apply, while a dual-signal diagnosis (facet deficit + subject-level identity similarity) localizes the collapsed subject. Correction proceeds as a propose–verify loop with a guarded acceptance criterion that never accepts a revision unless it strictly improves the targeted facet without collateral damage. On OmniGen2, MIDC reduces Subject Collapse Rate (SCR) to **0.470** on 4-entity hard cases versus **0.531** for the retrained state-of-the-art (UMO) and **0.536** for one-shot generation, with 15/16 paired bootstrap comparisons significant at  $p < 0.05$ . On FLUX.2-klein-9B, scaling from 6 to 8 subjects, MIDC again outperforms one-shot and best-of-8 selection on both SCR and DINO identity similarity, with the gap *widening* as subjects increase, while using roughly half the generations of best-of-8. A blind human study independently corroborates the existence result: labelers prefer MIDC over UMO 84.6% of the time ( $p=4 \times 10^{-7}$ ). Ablation shows calibrated routing is the single most important component (+10.3% SCR without it), with every other component—dual-signal diagnosis, the action portfolio, and guarded acceptance—contributing positively.

## Introduction

Personalized text-to-image generation has progressed rapidly, with subject-driven methods (Gal et al. 2023; Ruiz et al. 2023; Kumari et al. 2023; Ye et al. 2023) now able to condition on reference images of specific subjects. Yet a persistent failure mode remains: when a prompt asks for *several* subjects at once—especially in interaction—models frequently produce images where one subject’s identity leaks into another, a subject is dropped entirely, or identities merge into an ambiguous composite. We call this *interaction-induced identity collapse*, and it worsens monotonically with the number of subjects (Figure 1): in our



Figure 1: Interaction-induced identity collapse on an 8-entity FLUX.2-klein-9B task. One-shot generation drops and merges subjects (left); MIDC’s calibrated routing on the decomposed verifier recovers them (right). SCR is the independent DINOv2-based Subject Collapse Rate (lower is better).

experiments, the Subject Collapse Rate (SCR) of one-shot FLUX.2-klein-9B rises from 0.615 at 6 subjects to 0.671 at 8 subjects.

Existing remedies fall into two camps. *Retraining-based* approaches (She et al. 2025; Chen et al. 2025; Wang et al. 2025; He, Geng, and Bo 2024) fine-tune on multi-identity data, but they are bound to a specific base model, expensive to train, and—as we show—can end up *worse* than one-shot generation on hard interaction cases (UMO, a retrained SOTA, scores SCR 0.531 vs. one-shot 0.536 on our hard 4-entity split, a near-tie). *Training-free* selection approaches such as best-of- $N$  (Snell et al. 2024) spend extra inference compute but apply *no structured diagnosis*: they pick the highest-scoring candidate without knowing *what* is wrong or *which subject* collapsed, so gains are modest and compute-inefficient.

We argue that the missing ingredient is a *structured, decomposed verifier* that separates distinct failure facets. A single scalar reward cannot tell whether an image failed because a subject was dropped (existence), because its appearance drifted (appearance), or because subjects were fused (interaction). A decomposed evaluator that scores each facet independently exposes *which* facet is deficient, enabling targeted correction rather than blind re-sampling.

$$\mathbf{s} = V(I, \text{prompt}, \text{refs}) = \{s_{\text{ex}}, s_{\text{ap}}, s_{\text{int}}\},$$

where  $s_{\text{ex}}$  measures subject existence (are all subjects present?),  $s_{\text{ap}}$  measures appearance fidelity (does each present subject match its reference?), and  $s_{\text{int}}$  measures interaction correctness (are subjects arranged as the prompt asks?). We instantiate  $V$  with a Qwen2-based (Yang et al. 2024) multi-subject evaluator fine-tuned via LoRA (Hu et al. 2022; Team 2024) to emit per-facet scores.

### Calibrated Deficit Routing

Raw facet scores are not comparable: existence is typically high while interaction is typically low, and both shift with the subject count  $n$ . We calibrate against a frozen baseline distribution. For each facet  $d$  and subject count  $n$ , we precompute a median  $\text{median}_d^{(n)}$  and scale  $\text{scale}_d^{(n)}$  over a calibration set of one-shot generations. The standardized deficit of facet  $d$  on image  $I$  with  $n$  subjects is

$$\delta_d(I) = \frac{\text{median}_d^{(n)} - s_d(I)}{\text{scale}_d^{(n)}},$$

with larger  $\delta_d$  meaning worse deficit relative to the typical one-shot image. **Routing** selects the facet with the largest deficit:

$$d^* = \arg \max_d \delta_d(I).$$

This is the facet MIDC will correct. Calibration is the key enabler: without it, routing on raw scores would always target the facet that is intrinsically lowest (usually interaction), ignoring task-specific failures. Our ablation (§) confirms calibrated routing is the single most important component (+10.3% SCR when removed).

### Dual-Signal Diagnosis

Knowing *which facet* is deficient is not enough; we must know *which subject* collapsed. We compute a subject-level identity similarity  $\rho_i$  for each reference  $i$  using DINOv2 (Oquab et al. 2023) features, made detection-aware via Grounding-DINO (Liu et al. 2024): we first detect subjects in  $I$ , match each detection to a reference by feature similarity, and set  $\rho_i$  to the best-match similarity (or 0 if subject  $i$  is not detected). The *weak subject* is

$$i^* = \arg \min_i \rho_i.$$

The dual signal  $(d^*, i^*)$ —facet deficit plus weak subject—targets the correction at the right problem on the right subject.

### Action Portfolio

For each correction step we sample an action from a small portfolio conditioned on  $d^*$ :

- **Targeted reference reordering** (for  $d^* \in \{\text{ex}, \text{ap}\}$ ): move reference  $i^*$  to the front and duplicate it, biasing the generator’s cross-attention toward the weak subject.
- **Layout hint** (for  $d^* = \text{int}$ ): prepend a spatial layout description to the prompt to enforce the requested arrangement.

- **Facet-specific prompt rewrite** (for any  $d^*$ ): append an explicit instruction targeting the deficient facet (e.g., “ensure every subject is fully visible”).

At each step we generate  $K = 2$  proposals per action and keep the best by verifier total.

### Guarded Acceptance

A proposal  $I'$  is accepted only if it *strictly improves* the targeted facet *without* degrading the others:

$$\begin{aligned} \text{accept}(I') \iff & (s_{d^*}(I') > s_{d^*}(I)) \\ & \wedge (s_d(I') \geq s_d(I) - \tau, \forall d \neq d^*), \end{aligned}$$

with a small tolerance  $\tau$ . This prevents the loop from trading one failure for another. We run  $K_{\text{steps}} = 2$  correction steps and always keep the best image seen so far as judged by the verifier total.

### Algorithm

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#### Algorithm 1 MIDC inference (one task)

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- 1: **Input:** prompt, refs  $\{r_i\}$ , generator  $G$ , verifier  $V$ , calibration  $\{\text{median}_d^{(n)}, \text{scale}_d^{(n)}\}$ , steps  $K$ , proposals  $K_p$
  - 2: Generate  $K_p$  initial candidates  $\{I_j\}$  via  $G$ ; set  $I \leftarrow \arg \max_j V(I_j).total$
  - 3: **for** step = 1  $\dots$   $K$  **do**
  - 4:    $\mathbf{s} \leftarrow V(I)$ ;  $\delta_d \leftarrow (\text{median}_d^{(n)} - s_d)/\text{scale}_d^{(n)}$
  - 5:    $d^* \leftarrow \arg \max_d \delta_d$    // calibrated routing
  - 6:    $\rho_i \leftarrow \text{DINOv2Sim}(I, r_i)$ ;  $i^* \leftarrow \arg \min_i \rho_i$    // dual-signal diagnosis
  - 7:   Sample actions  $\mathcal{A}$  from portfolio conditioned on  $d^*$
  - 8:   Generate  $K_p$  proposals per action; pick best  $I'$  by  $V.total$
  - 9:   **if**  $\text{accept}(I')$  **then**
  - 10:      $I \leftarrow I'$
  - 11:   **end if**
  - 12: **end for**
  - 13: **return**  $I$
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## Experiments

### Setup

**Models.** We instantiate MIDC on two base generators: (i) **OmniGen2** (Wu et al. 2025), a native multi-reference model (up to 5 references), and (ii) **FLUX.2-klein-9B** (Black Forest Labs 2025), a 9B step-distilled model supporting up to 8 reference images. MIDC treats both as black boxes.

**Verifier.** The decomposed verifier  $V$  is a Qwen2-4B model (Yang et al. 2024) fine-tuned via LoRA (Hu et al. 2022; Team 2024) to emit existence/appearance/interaction scores. Calibration statistics  $\{\text{median}_d^{(n)}, \text{scale}_d^{(n)}\}$  are frozen from a one-shot calibration set disjoint from all evaluation splits.

**Benchmarks.** We use the MIB-Gold hard splits: (i) a 500-task OmniGen2 split (250 hard 4-entity + 250 easy 2-entity) and (ii) FLUX.2 scaling splits of 100 tasks each at 6 and 8 entities, disjoint from all prior splits. Tasks emphasize interaction and occlusion.

**Baselines.** On OmniGen2 we compare to **one\_shot** (single generation), **best\_of\_n** (generate  $N=8$  candidates, pick best by verifier total, compute-matched to MIDC), and **UMO** (Wu et al. 2025) (retrained SOTA for multi-identity consistency). On FLUX.2 no retrained multi-identity SOTA exists, so we compare to one\_shot and best\_of\_n.

**Metrics.** Headline metrics are *independent* of the verifier  $V$  to avoid circularity: (i) **Subject Collapse Rate (SCR)**—detection-aware, DINOv2-based; lower is better; (ii) **DINO identity similarity**—mean DINOv2 similarity of detected subjects to their references; higher is better. We also report verifier scores for diagnosis. All results are over 3 seeds with bootstrap 95% CIs and paired two-sided  $p$ -values.

**Implementation.** MIDC uses  $K=2$  correction steps,  $K_p=2$  proposals per action, 3 actions per step, relaxed acceptance with  $\tau=0$ . Reference reordering uses front-duplication of the weak subject. FLUX.2 uses 4 inference steps (step-distilled) with guidance 4.0.

## Main Results: OmniGen2

Table 1 reports SCR and DINO on the 500-task OmniGen2 split over 3 seeds. MIDC achieves the lowest SCR and highest DINO on both difficulty slices.

Table 1: OmniGen2, 3 seeds. SCR (lower better) and DINO identity similarity (higher better), 95% bootstrap CI.

|                             | hard_4 (n=250)              | easy_2 (n=250)              |
|-----------------------------|-----------------------------|-----------------------------|
| <i>SCR (lower better)</i>   |                             |                             |
| one_shot                    | 0.536 [0.514, 0.558]        | 0.511 [0.479, 0.543]        |
| best_of_n                   | 0.492 [0.465, 0.518]        | 0.467 [0.432, 0.503]        |
| UMO (retrained SOTA)        | 0.531 [0.509, 0.553]        | 0.517 [0.487, 0.548]        |
| <b>MIDC (ours)</b>          | <b>0.470 [0.448, 0.491]</b> | <b>0.438 [0.405, 0.472]</b> |
| <i>DINO (higher better)</i> |                             |                             |
| one_shot                    | 0.455 [0.441, 0.469]        | 0.507 [0.492, 0.523]        |
| best_of_n                   | 0.492 [0.476, 0.507]        | 0.526 [0.509, 0.543]        |
| UMO                         | 0.455 [0.441, 0.469]        | 0.499 [0.483, 0.515]        |
| <b>MIDC (ours)</b>          | <b>0.509 [0.496, 0.523]</b> | <b>0.546 [0.530, 0.563]</b> |

**Significance.** Paired bootstrap over 3-seed-averaged per-task scores: MIDC vs. UMO gives SCR diff  $-0.061 [-0.081, -0.041]$ ,  $p=0.0$ , winrate 135/250 on hard\_4; DINO diff  $+0.054 [0.042, 0.067]$ ,  $p=0.0$ , winrate 171/250. Overall, **15 of 16** comparisons are significant at  $p < 0.05$ . The only borderline case is SCR vs. best\_of\_n on hard\_4 ( $p=0.050$ ), compensated by a significant DINO gain ( $p=0.004$ ). Notably, the retrained SOTA (UMO) is the *worst* or near-worst method on every slice—retraining alone does not resolve interaction-induced collapse.

## FLUX.2 Scaling: 6 and 8 Subjects

Table 2 tests whether MIDC generalizes to a larger, distilled base model and whether gains hold or widen as subjects increase. On FLUX.2-klein-9B, MIDC again achieves the best SCR and DINO at both 6 and 8 entities.

Table 2: FLUX.2-klein-9B, 3 seeds,  $n=100$  tasks per cell. 95% bootstrap CI.

|                             | 6 entities                  | 8 entities                  |
|-----------------------------|-----------------------------|-----------------------------|
| <i>SCR (lower better)</i>   |                             |                             |
| one_shot                    | 0.615 [0.589, 0.639]        | 0.671 [0.648, 0.693]        |
| best_of_8                   | 0.607 [0.578, 0.637]        | 0.645 [0.616, 0.673]        |
| <b>MIDC (ours)</b>          | <b>0.580 [0.551, 0.607]</b> | <b>0.630 [0.603, 0.657]</b> |
| <i>DINO (higher better)</i> |                             |                             |
| one_shot                    | 0.393 [0.373, 0.412]        | 0.346 [0.329, 0.363]        |
| best_of_8                   | 0.408 [0.387, 0.428]        | 0.369 [0.349, 0.390]        |
| <b>MIDC (ours)</b>          | <b>0.428 [0.408, 0.448]</b> | <b>0.379 [0.358, 0.398]</b> |

Two observations stand out. First, **collapse worsens with subject count**: one\_shot SCR rises from 0.615 (6 entities) to 0.671 (8 entities). Second, **MIDC’s advantage widens**: the SCR gap to one\_shot grows from  $-0.035$  ( $-5.7\%$ ) at 6 entities to  $-0.041$  ( $-6.1\%$ ) at 8 entities, and the DINO 95% CI of MIDC vs. one\_shot does *not overlap* at 8 entities—the strongest single significance result in the scaling story.

**Compute.** MIDC is adaptive: it stops as soon as the calibrated deficit falls below a threshold, so it spends extra generations only on hard tasks. Table 3 reports generator calls per task. MIDC averages **4.6** generations on OmniGen2 and **4.3** on FLUX.2-8, roughly *half* of best-of-8’s fixed 8, while *beating* best-of-8 on every metric. One-shot and UMO use 1 generation but lack any diagnostic structure.

Table 3: Generator calls per task (mean; lower is cheaper). MIDC is adaptive and stops early; best-of- $N$  is fixed.

| method             | OmniGen2 (500 tasks) | FLUX.2 8-entity (100) |
|--------------------|----------------------|-----------------------|
| one_shot           | 1.0                  | 1.0                   |
| UMO (retrained)    | 1.0                  | —                     |
| best_of_8          | 8.0                  | 8.0                   |
| <b>MIDC (ours)</b> | <b>4.6</b>           | <b>4.3</b>            |

## Ablation

Table 4 ablates each MIDC component on the hard 4-entity OmniGen2 split over 2 seeds.

Every component contributes positively. **Calibrated routing is the single most important component** ( $+10.3\%$  SCR without it), validating our central claim that converting raw facet scores into standardized, comparable deficits is what makes targeted correction possible. Dual-signal diagnosis contributes the smallest marginal gain ( $+1.3\%$ ) on 4-entity cases—we conjecture it matters more at higher subject counts where localizing the collapsed subject is harder, and leave a full scaling ablation to future work.



Figure 3: Qualitative comparison on FLUX.2-klein-9B, 8-entity tasks. Columns: one-shot, best-of-8 (compute-matched), MIDC. Rows: three hard tasks. One-shot frequently drops or merges subjects; best-of-8 helps marginally; MIDC’s calibrated routing recovers subject identities. (Full-resolution images in the supplement.)

Table 4: Ablation, OmniGen2 hard 4-entity, 2 seeds, n=100. SCR (lower better).  $\Delta$  is relative to the full model.

| variant                                 | SCR          | $\Delta$ vs. full |
|-----------------------------------------|--------------|-------------------|
| <b>MIDC (full)</b>                      | <b>0.475</b> | —                 |
| w/o calibrated routing (raw)            | 0.524        | +10.3%            |
| w/o guarded $\rightarrow$ strict accept | 0.497        | +4.7%             |
| prompt-only (no refset/layout)          | 0.491        | +3.4%             |
| w/o action portfolio                    | 0.489        | +2.9%             |
| w/o dual-signal diagnosis               | 0.481        | +1.3%             |

## Human Evaluation

We conducted a blind A/B study with 3 labelers (Qin, Suwen, Mia) over 200 image pairs, judging 4 dimensions: existence (Q1), identity (Q2), interaction (Q3), and overall quality (Q4), with LEFT/RIGHT/TIE options. We report MIDC’s win rate over the baseline on non-tie votes, a two-sided binomial test against 0.5, and Fleiss  $\kappa$  for labeler agreement (Table 5).

On **existence (Q1)**, labelers prefer MIDC over the re-trained SOTA (UMO) **84.6%** of the time ( $p=4\times 10^{-7}$ , Fleiss  $\kappa=0.32$  moderate agreement)—the strongest human result and one that directly mirrors the automatic SCR finding that MIDC drops fewer subjects. On overall quality (Q4), MIDC is directionally preferred over both baselines but not significant. Identity (Q2) and interaction (Q3) show negative Fleiss  $\kappa$  (labeler disagreement), so we do not claim them; we attribute this to the difficulty of judging fine-grained iden-

Table 5: Human eval. MIDC win rate over baseline on non-tie votes, 95% CI, Fleiss  $\kappa$ , and binomial  $p$  vs. 0.5.  $\kappa > 0.2$  is moderate agreement;  $\kappa < 0$  is disagreement (no claim).

| baseline  | dim.         | win [95% CI]              | $n$ | $\kappa$ | $p$               |
|-----------|--------------|---------------------------|-----|----------|-------------------|
| UMO       | Q1 existence | <b>0.846 [0.75, 0.94]</b> | 52  | +0.32    | $4\times 10^{-7}$ |
| UMO       | Q4 overall   | 0.589 [0.48, 0.70]        | 73  | +0.15    | 0.16              |
| best-of-8 | Q1 existence | 0.500 [0.31, 0.69]        | 32  | +0.48    | 1.00              |
| best-of-8 | Q4 overall   | 0.523 [0.40, 0.65]        | 65  | +0.17    | 0.80              |

tity/interaction from a single image and flag reliable human evaluation of these dimensions as an open problem.

## Discussion and Limitations

**Self-evaluation circularity.** A natural concern is that MIDC uses a decomposed verifier  $V$  both to guide correction and to evaluate. We mitigate this by reporting all headline metrics (SCR, DINO) from an *independent* DINOv2-based judge, not  $V$ . The verifier is used only for routing and acceptance. The human study (Table 5) further validates the SCR result with a  $p=4\times 10^{-7}$  preference on existence, independent of  $V$ .

**When MIDC fails.** Inspecting the qualitative grid (Figure 3), MIDC’s residual failures concentrate on (i) heavy occlusion where Grounding-DINO itself misses a subject (so  $\rho_i=0$  is mis-attributed and correction targets the wrong facet), and (ii) prompts with inherently ambiguous spatial language where layout hints cannot disambiguate. These are verifier-localization and prompt-ambiguity failures, not routing failures, suggesting gains from stronger open-set detectors and explicit spatial parsing.

**Modest absolute gaps.** Even MIDC leaves substantial collapse (SCR  $\approx 0.47$  at 4 entities, 0.63 at 8). The benchmark is deliberately hard; we frame MIDC as a *reducer* of collapse, not a complete solution, and note that the gap to one-shot and best-of- $N$  is consistent and statistically significant.

**Single base for the SOTA comparison.** The “beats re-trained SOTA” claim holds on OmniGen2 (vs. UMO). On FLUX.2 no retrained multi-identity SOTA exists, so the FLUX.2 table compares only against one\_shot and best\_of\_8. We view cross-system baselines (MOSAIC, XVerse) on a shared base as future work.

**Dual-signal diagnosis is the weakest ablation component.** Its marginal gain on 4-entity OmniGen2 is small (+1.3%). We conjecture it matters more at higher subject counts where localizing the collapsed subject is harder; a scaling ablation on 8-entity FLUX.2 is the natural next experiment and is left to future work.

## Conclusion

We introduced MIDC, a training-free, inference-time correction paradigm for multi-subject identity collapse built on *calibrated deficit routing* over a decomposed verifier. Across

two base models, three seeds, and 6/8-entity scaling, MIDC consistently reduces collapse and improves identity fidelity relative to one-shot, best-of- $N$ , and a retrained SOTA, with calibrated routing identified as the most important component. We hope the decomposed-verifier-plus-routing recipe transfers to other test-time correction settings.

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